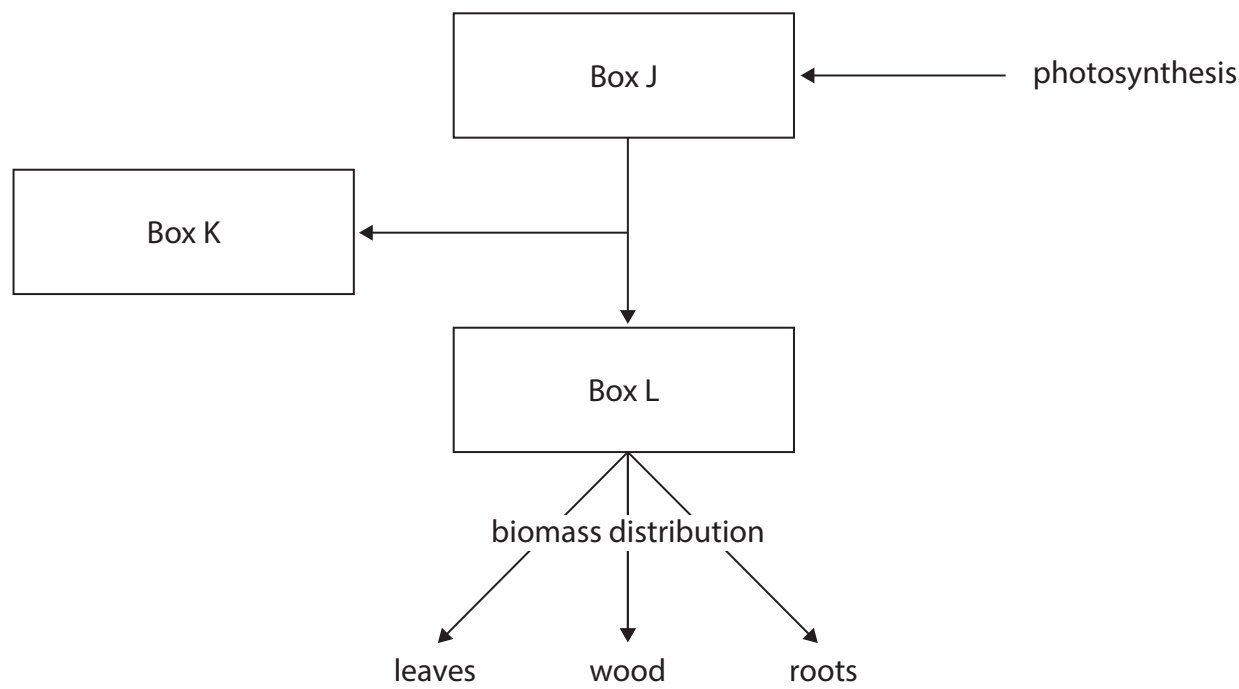


1 The diagram shows some processes linking photosynthesis to productivity and the distribution of biomass in the trees in a forest.



(a) Which row of the table identifies the box that represents net primary productivity (NPP), the box that represents gross primary productivity (GPP) and the box that represents respiration (R)?

(1)

	Box J	Box K	Box L
☐ A	NPP	R	GPP
☐ B	GPP	NPP	R
☐ C	GPP	R	NPP
☐ D	R	GPP	NPP

(b) Which are suitable units for NPP?

(1)

- ☐ **A** kJ m yr^{-1}
- ☐ **B** $\text{kJ m}^{-1} \text{yr}^{-2}$
- ☐ **C** $\text{kJ m}^{-2} \text{yr}^2$
- ☐ **D** $\text{kJ m}^{-2} \text{yr}^{-1}$

(c) The biomass distributed to the leaves, wood and roots of the trees in this forest was 12 500 a.u.

- (i) The biomass was distributed 34% to the leaves, 37% to the roots and the rest to the wood of the trees.

Calculate how many units of biomass were distributed to the wood of these trees.

(1)

Answer a.u.

- (ii) Which row of the table shows what can be found in the biomass of leaves and roots of trees?

(1)

	Leaves	Roots
☐ A	chlorophyll and starch	chlorophyll and protein
☐ B	chlorophyll and glycogen	lignin and protein
☐ C	chlorophyll and starch	starch and protein
☐ D	glycogen and cellulose	cellulose and protein

(d) Forests are examples of ecosystems.

State what is meant by the term **ecosystem**.

(1)

(Total for Question 1 = 5 marks)
